

Question

Write a Java program to demonstrate Single Inheritance using constructors, method overriding and method overloading. Calculate Simple Interest.

Steps

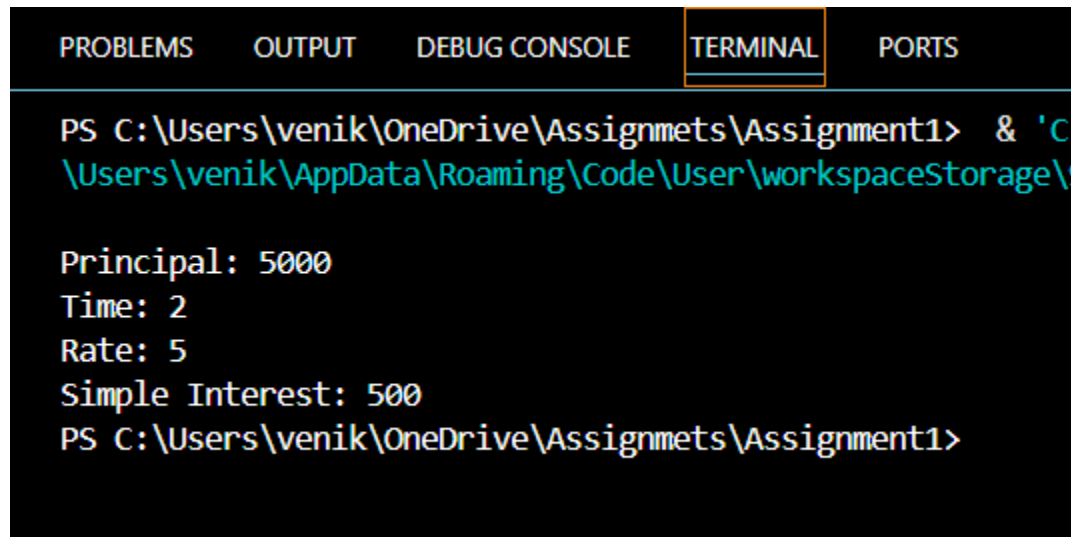
1. Create a parent class `Interest`.
2. Declare principal and time.
3. Create a constructor to initialize them.
4. Create a `calculate()` method in the parent class.
5. Create a child class `SimpleInterest` extending `Interest`.
6. Add rate of interest in the child class.
7. Create a constructor in the child class.
8. Use `super()` to call the parent constructor.
9. Override the `calculate()` method.
10. Overload the `calculate()` method with different parameters.
11. Create an object in `main()`.
12. Call the methods and display Simple Interest.

Algorithm

START

1. Create the parent class `Interest`.
2. Declare principal `P` and time `T`.
3. Initialize `P` and `T` using constructor.
4. Define `calculate()` method.
5. Create child class `SimpleInterest` extending `Interest`.
6. Declare rate `R`.
7. Initialize `R` using child constructor.
8. Call parent constructor using `super()`.
9. Override `calculate()`.
10. Calculate $SI = (P \times T \times R) / 100$.

11. Overload calculate() with different parameters.
12. Create SimpleInterest object.
13. Call the methods.
14. Display Simple Interest.



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\venik\OneDrive\Assignmets\Assignment1> & 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\9\
Principal: 5000
Time: 2
Rate: 5
Simple Interest: 500
PS C:\Users\venik\OneDrive\Assignmets\Assignment1>
```